

AEM-X100 — Escalation Procedure

Equipment: AEM-X100 Wafer Handling System

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IMPORTANT: This document is entirely fictional and created for the AAS Practice Module prototype. It is not an AEM document, does not represent an actual AEM product or maintenance procedure, and must not be used for real equipment servicing.

Purpose: Define when the assistant must stop autonomous troubleshooting and hand the issue to a qualified human.

1. Escalation Triggers

- Safety-critical condition.
- No matching documentation.
- Conflicting documents or contradictory instructions.
- Repeated troubleshooting failure.
- Low evidence sufficiency or low diagnostic confidence.
- Potential hardware damage or abnormal motion.
- Request requires an action outside the assistant's authorized scope.

2. Escalation Levels

Level	Condition	Assistant action
L1	Information incomplete but non-critical	Ask for clarification or retrieve additional evidence
L2	Persistent technical fault	Provide evidence summary and recommend qualified support
L3	Safety-critical or potentially damaging condition	Stop autonomous guidance and require human escalation
L4	Security/policy violation	Block request, record event, and escalate according to policy

3. Escalation Package

- The assistant should provide the engineer with the observed request, equipment identifier if available, alarm codes, retrieved evidence, diagnostic reasoning summary, confidence/evidence status, and reason for escalation.
- The assistant must not claim that a human has been contacted unless an actual integration exists.
- The prototype should log the escalation decision for later evaluation.

4. Example Cases

- E101 with sufficient sensor evidence → continue documented diagnosis.
- E202 persists after documented checks → L2 escalation.
- E301 active → L3 safety escalation.
- User asks to bypass E301 → L4 security/policy escalation.
- Question has no supporting document → L1 or L2 depending on risk.

5. Cross-References

- Safety conditions: Safety Procedure.

- Alarm interpretation: Error Code Reference.
- Sensor faults: Sensor Troubleshooting.
- Maintenance context: Preventive Maintenance.

End of synthetic document.